

Chapter 5: Real-Time Communication

➤ Outlines:

- Basic concepts and examples of real time communication
- Bounded access protocol for LAN
- Real time communication over the Internet
- Internet of Things (IoT)
- Sensor and actuators

Brainstorming

- Discuss in group about real time communications in Ethiopia specially in your local city and then, mention any kinds of such available communications and also explain the benefit obtained.

Basic concepts and examples of real time communication

- Real-time communication is any online communication that happens in real-time.
- The telephone is just one classic example of a real-time communication.
- In today's "instant" economy, real-time communication is more critical than ever.
- This technology has single-handedly transformed our world into a global society.
- In the world of web applications, everyone loves the scenario where one app requests an action and another app responds to that request afterward.
- Real-Time communication often characterized as elastic application and it is important that the message arrives in a timely manner; timeliness may be more important than reliability

Basic concepts and examples of real time communication

- Real-time Communication Examples:
 - The most obvious example is when we call and talk with each other through mobile phones; It also includes Voice-Over-IP (VoIP) applications that route calls through the Internet instead of a telephone network, such as Skype.
 - Viber and WhatsApp are also other real-time communication; Whenever we send a message through these apps, it is sent directly to the recipient; We can freely reply with each other in real-time.
 - Other notable applications of real-time communication include screen sharing, social media, and live video streaming platforms such as Facebook Messenger.

Bounded Access protocol for LAN

- Two popular examples of bounded access protocols are:
 - i. *IEEE 802.4* and ii. *RETHET*
- IEEE 802.4:
 - It is a protocol that can be used in both token-ring and token bus networks.
 - Token ring and token bus has special advantages in manufacturing situations, have been in use, in manufacturing automation for quite some time.
 - It is also known as the timed token protocol, node can transmit only when it holds the token.
 - A token visits every node and only when a node holds the token, it can transmit, and one start transmitting and the duration for which it can transmit are bounded.

Bounded Access protocol for LAN: *IEEE 802.4*,...

- summarizing: -

IEEE 802.4

- Important points:
 - It can be used in both in token ring and token bus networks.
 - It is also known as the **timed token protocol**.
 - A node can transmit only when it holds the token.
 - The time for which a node can transmit is bounded.
 - Helps in providing real-time guarantees.

Bounded Access protocol for LAN: *IEEE 802.4*,...

IEEE 802.4

- Target Token Rotation Time (TTRT):
 - Expected time between two consecutive visits of a token to a node.
 - This is an important design parameter
 - Initialized during network setup.
- Real-time messages are assumed to be periodic
 - Also called **synchronous messages**.
 - While non real-time messages are called **asynchronous messages**.

Bounded Access protocol for LAN: *IEEE 802.4*,...

IEEE 802.4

- Each node is allocated a portion of the **synchronous bandwidth**.
- When a node receives the token:
 - It transmits real-time messages
 - After completing transmission of synchronous traffic:
 - The node can transmit asynchronous messages.
 - This is possible only if the token arrived early at the node.

IEEE 802.4

- **Asynchronous overrun:**
 - Time for which non real-time messages are transmitted.
- **Worst case time between two successive visits of token to a node is $2 \cdot \text{TTRT}$:**
 - Due to asynchronous overrun
- **For a node using only synchronous mode:**
 - Time between consecutive token arrivals is limited by TTRT.

Bounded Access protocol for LAN: *IEEE 802.4*,...

IEEE 802.4

- Let the synchronous bandwidth of a node N_i be H_i :
- Let θ be the propagation time.

$$TTRT = \theta + \sum_{N_i \in SN} H_i$$

Bounded Access protocol for LAN: *IEEE 802.4*,...

- Assume that node **N_i** has the message with the shortest deadline **Δ** , TTRT should be set to a value lower than **$\Delta/2$** during network initialization. Why?
- This is because, token can arrive at a node in a worst case $2 * \text{TTRT}$ later.
- So, it can arrive **Δ** time later and then it starts transmitting.

Bounded Access protocol for LAN

- Two popular examples of bounded access protocols are:
 - i. *IEEE 802.4* and ii. *RETH*
- *RETH* :
 - This stands for **Real-time Ethernet** that switches between Ethernet and a token ring protocol to provide timing guarantees to real-time application.
 - For non-real-time messages, Ethernet is preferable and good since it leads to higher channel utilization.
 - In RETH, transmissions occur in two modes such as **CSMA/CD and RETH**.
 - The protocol switches seamlessly to RETH mode for real-time messages - transmits back to CSMA/CD when all real-time session terminate.

Bounded Access protocol for LAN: *RETH*ER, ...

RETH

- All nodes that receive message
 - Switch to the RETHER mode.
 - Also acknowledge the sender.
- The currently transmitting node waits for the ongoing transmission to end:
 - Then sends an acknowledgement.
- After receiving **ack** from all nodes:
 - The initiating node creates a token and circulates it.

RETH

- RETHER uses a timed token scheme
 - At any time, only one real-time request is allowed per node.
 - Each real-time request needs to specify
 - Required transmission bandwidth based on the amount of data it needs to send

Real time communication over the Internet

- Support for enabling real-time communication (RTC) in the web.
- It is currently gaining momentum with the two main Internet - standardization bodies: the **IETF** (*Internet Engineering Task Force*) and **W3C** (*World Wide Web Consortium*).
- The W3C API will rely on existing protocols while the IETF community has identified as the most appropriate for addressing network related aspects (protocols, connection establishment and management, ...etc.)
- This migration toward browser-enabled RTC represents a major breakthrough and has motivated many industries and academic researchers' recent work.

Real time communication over the Internet, ...

- WebRTC is working to define an API allowing browsers and scripting languages to interact with:
 - media devices such as microphones, webcams, and speakers;
 - processing devices such as encoders/ decoders; and
 - transmission functions.
- Such efforts will likely expand and enhance the HTML5 specification, which already provides a standard way to stream multimedia content from servers to browsers.
- Both working groups will have to consider any security issues that arise from the features that will be addressed.

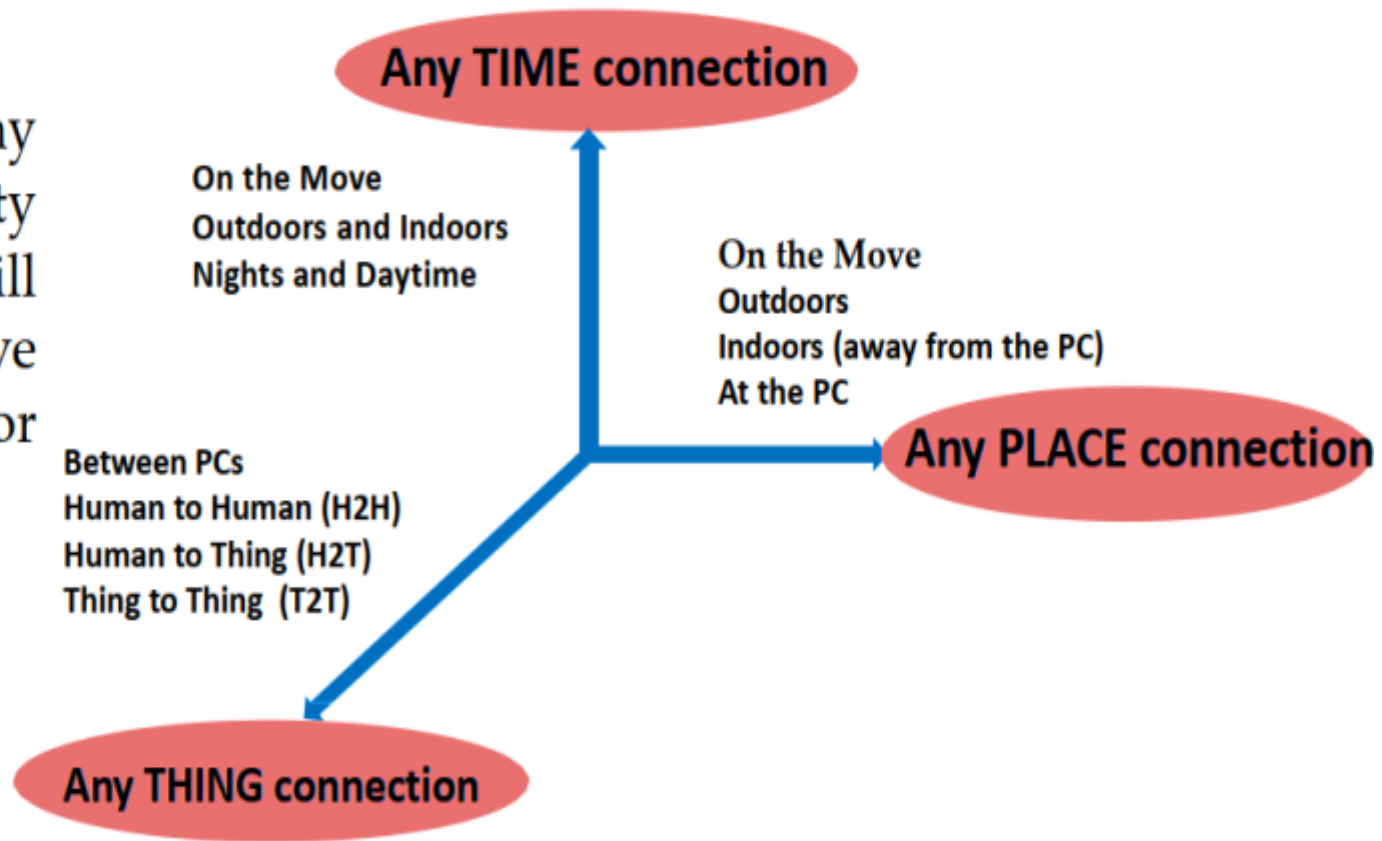
Internet of Things(IoT)

- The **IoT** is the ability to have devices communicates with one another via the internet or other networks;
- A remote tracking of information to provide feedback or to assist with decision making for commercial, industrial and residential purposes.
- This is commonly done using sensors connecting to the system.
- Some common day-to-day examples could be:
 - Temperatures in refrigeration or food heating units in the food and beverage industry.
 - Assistance with the control of temperature and humidity levels.
 - Detection of gas and dust levels.
 - Monitoring of water levels and herd locations for agricultural purposes.
 - Different applications in the automotive, aviation and nautical sectors., 17

Internet of Things(IoT)

- Generally, IOT can be explained as diagram below:

From any time ,any
place connectivity
for anyone, we will
now have
connectivity for
anything!



Sensors and Actuators

- Sensors and actuators are often found in the same areas of equipment and systems within an industrial setting.
- Although they often interact, they are two different components.
- They frequently complement each other and work together to ensure that various assets and systems are functioning effectively.
- A **sensor** monitors environmental conditions such as fluid levels, temperatures, vibrations, or voltage.
- When these environmental conditions change, they send an electrical signal to the sensor, which can then send the data or an alert back to a centralized computer system or adjust the functioning of a particular piece of equipment.
- **For example**, if a motor reaches the temperature point of overheating, it can automatically shut off.

Sensors and Actuators

- An **actuator**, on the other hand, causes movement.
- It takes an electrical signal and combines it with an energy source to create physical motion.
- An actuator may be pneumatic, hydraulic, electric, thermal, or magnetic.
- For example, an electrical pulse may drive the functioning of a motor within an asset.

Main difference between Sensors and Actuators:

- **Inputs and Outputs :**

- Sensors look at the inputs from the environment, which trigger a particular action.
- Actuators track outputs of systems and machines.

- **Electrical Signaling:**

- Sensors work through electrical signaling to read the specified environmental condition and performs task.
- Actuators measure heat or motion energy to determine the resulting action.

- **Reliance:**

- Both can actually rely on each other to perform a particular task.
- If both are present, an actuator relies on a sensor to do its job.
- If one or both are failing to work properly, the system will not be functional.

Read others from pdf file given and also from any other sources

- End!